FlowMLLab - Week 1.2

Pressure-Velocity Coupling in a Lid-Driven Cavity

SIMPLE, PISO and PIMPLE
A shared finite-volume Python implementation

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The central question

How does pressure enforce incompressibility, and how can different coupling loops reach the same flow with different computational costs?

Initial qualified case: Re=100. Two uniform grids. Executed fields, benchmark comparisons, momentum and continuity checks. An independent extension; Week 1 remains unchanged.

1. One physical problem, two formulations

Governing equations

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}^T) = -\nabla p + \frac{1}{\mathrm{Re}} \nabla^2 \mathbf{u}$$

$$\nabla \cdot \mathbf{u} = 0$$

Use a unit square, unit lid speed and density, and viscosity 0.01. All walls are impermeable; tangential velocities are zero except for the moving lid.

Week 1 solves streamfunction and vorticity using the relations below. Pressure is recovered separately from steady momentum. Week 1.2 solves velocity and pressure together.

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x}, \qquad \nabla^2 \psi = -\omega$$

Why add this extension?

Students can inspect pressure as the continuity constraint, shared-face mass budgets, under-relaxation, and physical-time versus coupling iterations. The pressure-velocity formulation also provides a route toward three-dimensional solvers.

Learning outcomes: derive the pressure operator; distinguish the three loops; verify mass and gauge invariance; compare measured errors and cost; state the limits of a steady benchmark.

2. Shared MAC finite-volume operators

Pressure p is stored at cell centers. Horizontal velocity u lies on vertical faces; vertical velocity v lies on horizontal faces. Arrays use y-first, x-second indexing.

Momentum and pressure algebra

$$Aq = b - Gp, \qquad d_i = \frac{1}{A_{ii}}$$

$$H(q) = b - \text{offdiag}(A)q, \qquad q = d(H - Gp)$$

$$L = -DdG, \qquad Lp = -D(dH)$$

Here d denotes the diagonal operator with entries d_i. D and G are the shared discrete divergence and pressure-gradient operators.

D computes the net velocity flux divided by cell volume. A shared face contributes with opposite signs to its two neighbors. This gives exact interior flux cancellation.

Normal wall velocities are zero. Tangential wall diffusion uses half-cell distances. The lid-corner discontinuity is retained. Central convection and diffusion use the same frozen advecting face velocities for every coupling method.

Gauge and compatibility

L has a constant nullspace. The closed-cavity RHS must sum to zero. Pin one cell during the sparse solve; shift reported pressure to zero mean. A pressure constant cannot change any velocity correction.

Sparse LU is a linear solver, independent of the pressure-coupling algorithm. No multigrid or packaged CFD solver is used.

3. SIMPLE: steady correction and relaxation

1. Build and under-relax steady momentum using the current velocities.
2. Predict face velocities from the current pressure.
3. Solve for the pressure correction.
4. Correct face velocities fully.
5. Relax pressure; repeat until momentum and continuity pass.

Predict, enforce continuity, and update

$$Aq^{*} = b - Gp^{k}, \qquad Lp' = -Dq^{*}$$

$$q^{k+1} = q^{*} - dGp'$$

$$p^{k+1} = p^{k} + \alpha_p p'$$

$$\alpha_u = 0.7, \qquad \alpha_p = 0.3, \qquad \|r_{\text{mom}}\|_{\infty} < 10^{-7}$$

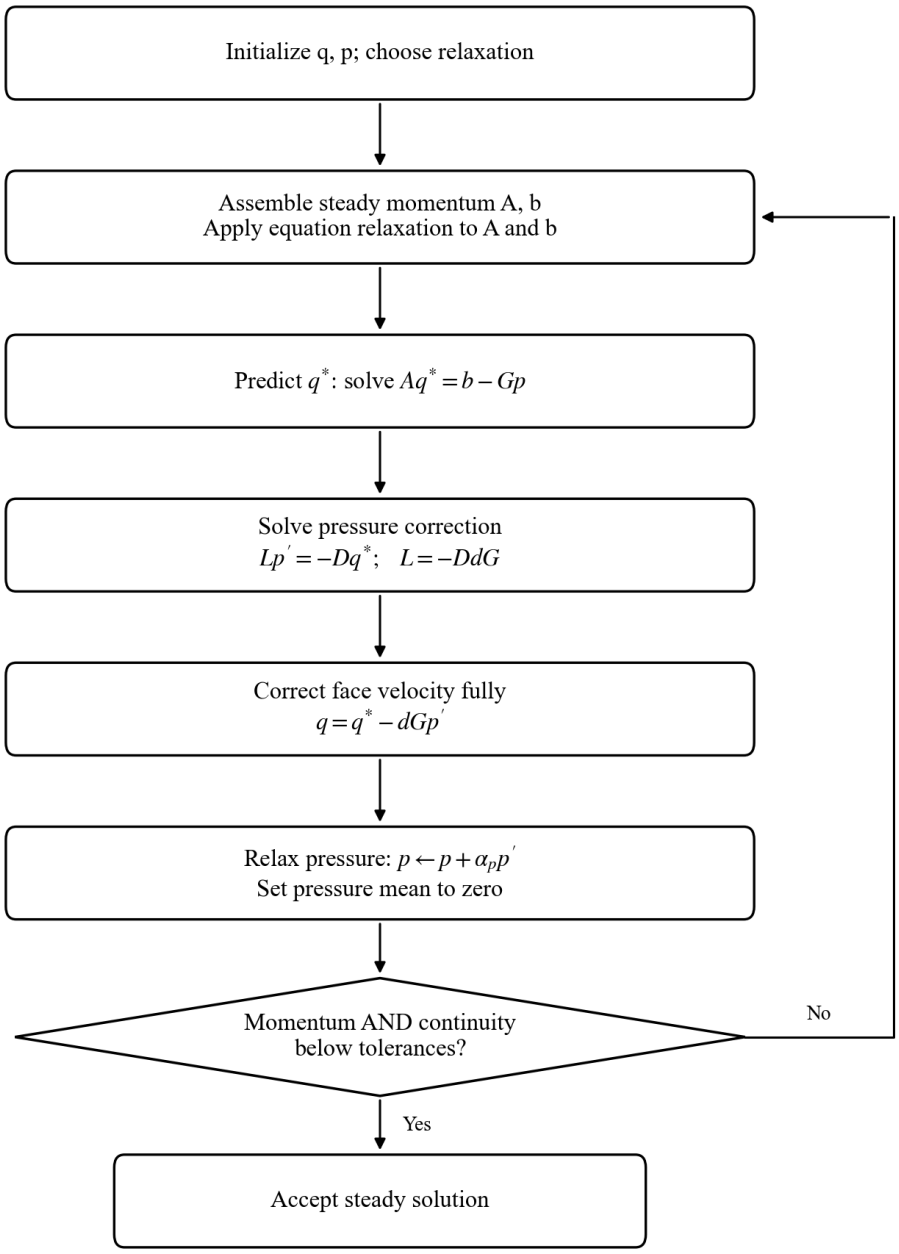
Advantage: transparent steady pressure coupling and useful relaxation experiments.
Limitation: sensitivity to relaxation; iteration count is not physical time or a fair cost measure.

Worked mass correction

If a tentative field has a nonzero cell divergence, pressure correction changes shared face fluxes until the corrected net flux vanishes. Independent velocity clipping would generally destroy that conservation property.

3.1 SIMPLE control flow

SIMPLE: a steady coupling iteration



The iteration index is not physical time. $d_i = 1/A_{ii}$.

Equation relaxation modifies A and b; pressure relaxation modifies the pressure update. The conservative velocity correction is applied fully. Reassemble the next iteration using corrected face velocities. Monitor the unrelaxed momentum equations, not only the update size.

4. PISO: corrections inside a physical step

1. Keep old-time velocities fixed.
2. Assemble and solve backward-Euler momentum for tentative velocity.
3. Recompute the off-diagonal momentum contribution H.
4. Solve absolute pressure and correct velocity using the relations below.
5. Repeat the pressure/velocity correction with the updated H.

Pressure and velocity correction

$$H^{(m)} = b - \mathrm{offdiag}(A)q^{(m)}$$

$$Lp^{(m+1)} = -D\left(dH^{(m)}\right)$$

$$q^{(m+1)} = d\left(H^{(m)} - Gp^{(m+1)}\right)$$

Recorded settings: two pressure corrections per physical step, with $\Delta t=0.05$. The old-time source remains fixed within both corrections.

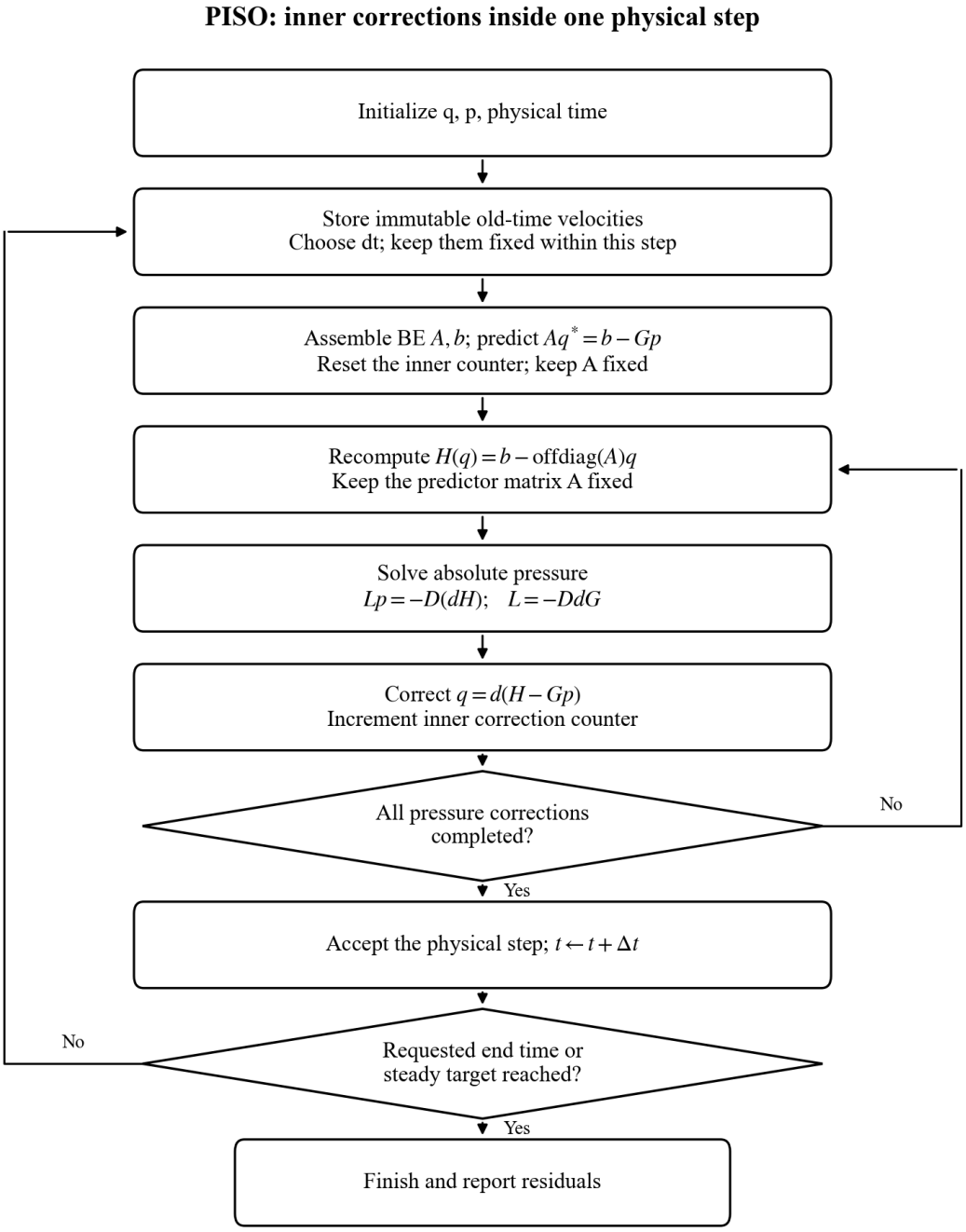
Advantage: additional pressure-momentum consistency without a complete nonlinear outer loop.
Limitation: more pressure solves per step; temporal accuracy still depends on Δt and the time discretization.

Do not mislabel a solver

Two Jacobi/LU iterations on a single pressure system are not two PISO corrections. The extra correction must update the momentum coupling using the corrected velocity.

Backward Euler has first-order temporal accuracy. Marching to steady state does not validate startup trajectories.

4.1 PISO control flow



An inner correction updates H; it is not a linear-solver sweep.

The second correction changes H through neighbor velocities while keeping A fixed. This improves coupling with the frozen predictor equations. It cannot fully remove error from lagged nonlinear convection coefficients. The old-time source must not change inside the correction loop.

5. PIMPLE: outer momentum iteration plus PISO

Inside one physical time step, repeat momentum assembly and prediction, then the PISO correction loop. The new velocity changes nonlinear convection coefficients for the next outer loop.

Recorded settings

Three outer loops; two pressure corrections in each; $\Delta t=0.05$. Old-time velocities remain immutable across all six pressure corrections.

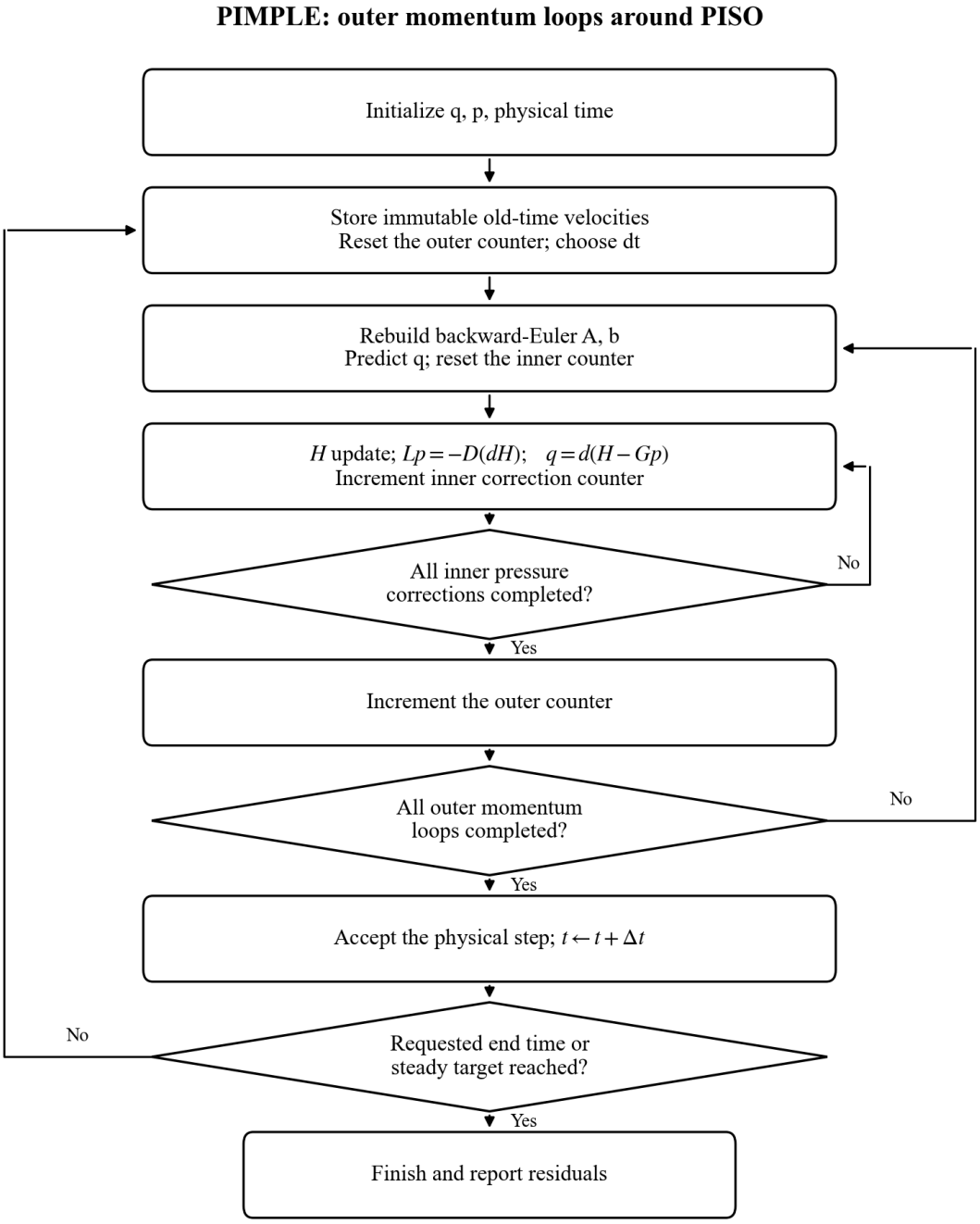
Advantage: additional nonlinear consistency within each time step; useful when a single PISO pass is insufficient.
Limitation: repeated momentum and pressure solves increase cost. Larger stable steps do not guarantee accurate transient dynamics.

An exact implementation check

Set the number of outer loops to one. With identical settings, the resulting PIMPLE step must match this PISO step exactly.

For this simple steady Re=100 cavity, extra outer loops need not be economical. A time-dependent lid and a matched temporal-error target are better tests of potential PIMPLE benefits.

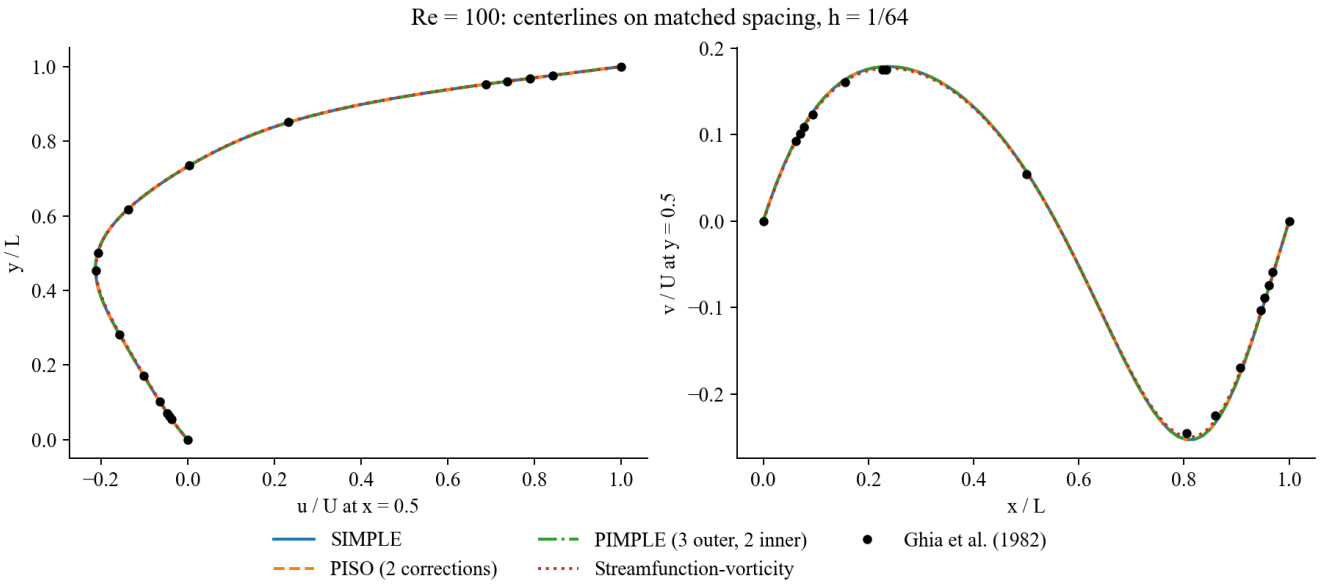
5.1 PIMPLE control flow



Old-time fields stay fixed in BOTH loops. One outer loop = PISO.

The inner loop updates pressure and velocity for a fixed predictor matrix. The outer loop rebuilds that matrix and solves momentum again. This release uses fixed counts, not adaptive residual-based loops. One outer loop gives the implemented PISO limit.

6. Benchmark velocity and two-grid evidence



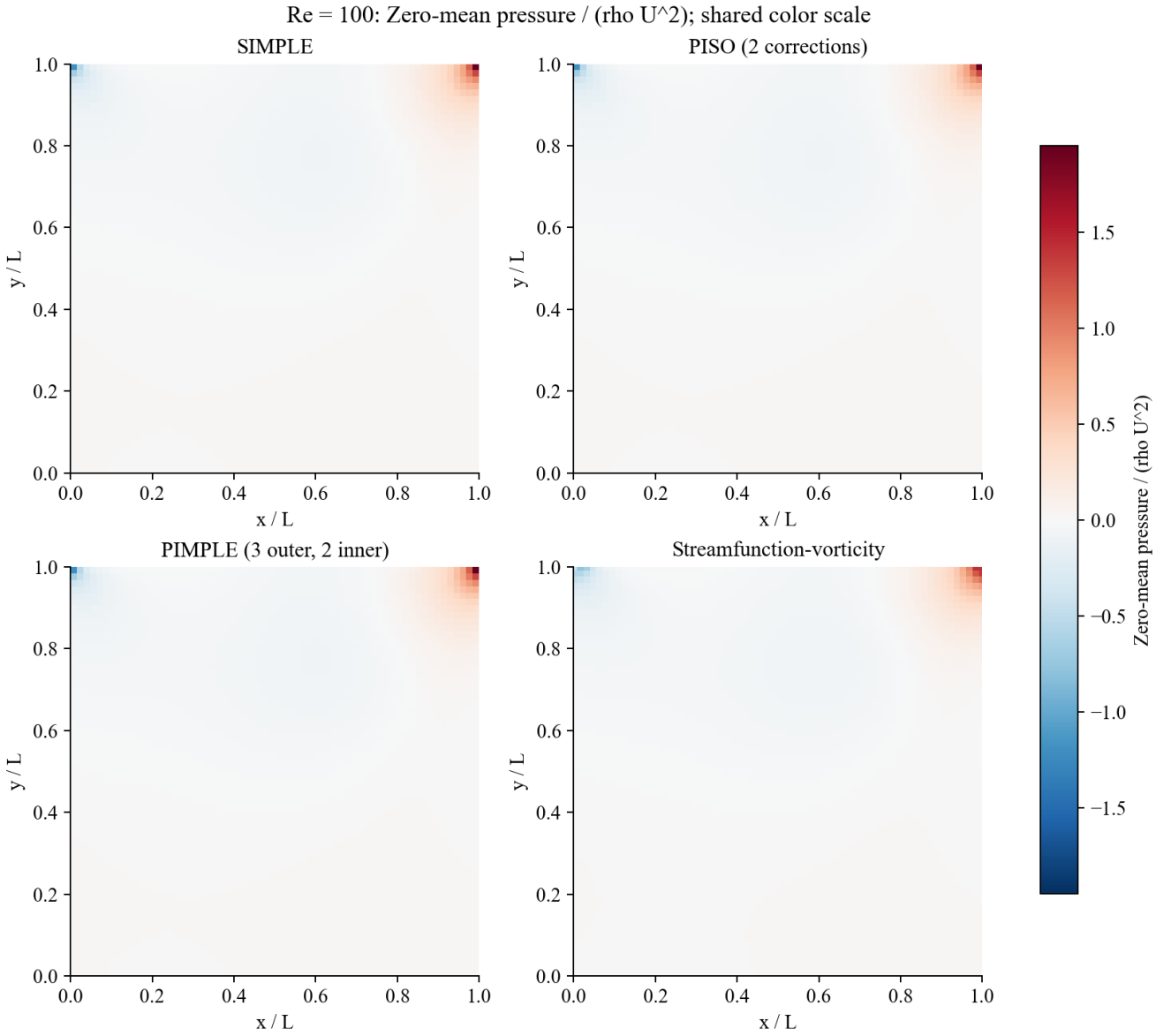
Ghia, Ghia and Shin (1982) velocity markers are compared at their stated coordinates, including wall endpoints. MAC face locations and FD nodes are interpolated only for the declared centerline diagnostics.

Fair resolution comparison

32/64 FV cells per side correspond to 33/65 FD nodes: compare h, not array length. The three FV algorithms use identical spatial equations. The independent FD formulation is a cross-check with a different truncation error.

The FD reference errors decrease on refinement; FV errors against Ghia increase slightly between these two grids. This nonmonotone trend is retained explicitly. Coarse-grid agreement with discrete benchmark data does not establish continuum accuracy. Further refinement and another reference are needed; no asymptotic order is claimed. Ghia velocity tables do not validate pressure.

7. Physical fields and pressure interpretation

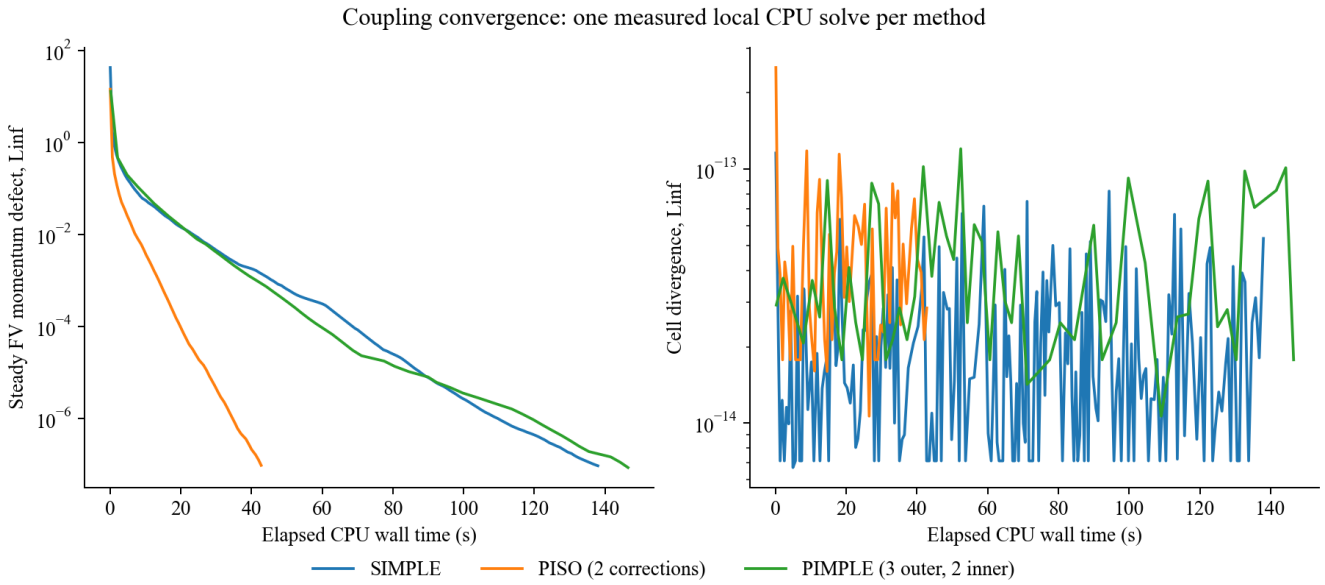


All pressure panels use the same zero-mean convention and common color limits. The three converged FV fields agree closely; the FD recovered pressure can differ near the singular lid corners.

Pressure evidence

Comparing pressure between independent formulations is informative but is not an exact-pressure validation. The notebook also shows speed, streamlines and vorticity on shared physical scales.

8. Convergence, cost and stopping conditions



The plotted steady FV momentum defect has units U^2/L, with $U=L=1$. Continuity is the maximum cell divergence. The elapsed-time axis includes actual numerical work, so differences in loop cost remain visible.

The FD reference uses a vorticity-update stopping norm and is not overlaid as if it were the same FV residual. A small under-relaxed update is not a substitute for a momentum-equation residual.

Measured conclusion

All six new FV runs reach the declared steady tolerance. On each grid, the three methods agree within 2×10^{-6} in u, v and zero-mean p. Timings are one local CPU solve per method, not a universal speed ranking.

9. What comparative studies actually show

Different studies answer different questions. None of these sources is a controlled comparison of our three Python implementations. There is no universal fastest coupling method.

Issa (1986): the transient motivation

PISO approximates the implicitly discretized equations through staged pressure/velocity corrections. Its companion study reports lower effort for time-evolving flows at sufficiently small splitting error. These publisher abstracts do not justify arbitrary large steps or universal steady superiority.

JCP 62, 40-65; companion paper, 66-82.

Pascau and Garcia (2010): ranking depends on the case

Their collocated-grid cavity study favored standard PISO among the tested variants at Re=1000. At Re=5000, a SIMPLEC-based variant performed better over a broader relaxation range. This is not a PIMPLE test and does not transfer as a MAC-grid ranking.

Enhancement of PISO Scheme in Collocated Grids, ECCOMAS CFD 2010.

Brogi et al. (2024): coupling and time-step accuracy

A modern OpenFOAM precision study discusses PISO/PIMPLE time-step choices in a compressible shock tube. Extra coupling can make larger steps useful, but increasing dt still affects accuracy. This is not an incompressible-cavity speed comparison.

On floating point precision in CFD using OpenFOAM, FGCS 152, 1-16. The 2022 preprint is an earlier version with different settings.

Our assessment rule: compare measured cost at a specified field-error and equation-defect target; retain failures. An iteration count alone is insufficient.

10. Higher Reynolds number: separate the causes

There is no universal Reynolds-number limit of SIMPLE, PISO or PIMPLE. Increasing Re can change spatial resolution requirements, nonlinear coupling and the physical/model behavior simultaneously.

A. Spatial convection and mesh resolution

$$a_E = \frac{\nu}{h^2} - \frac{u_E}{2h}, \qquad \mathrm{Pe}_h = \frac{|u_E| h}{\nu}$$

$$\mathrm{Pe}_h \le 2 \implies a_E \ge 0$$

The displayed bound is a sufficient positivity condition for this central neighbor coefficient, not a universal solver-failure threshold. Near unit advective speed, Re/N is about 15.6 for Re=1000 on 64 cells. Pressure corrections alone cannot resolve missing boundary-layer or vortex scales.

B. Nonlinear equation consistency

Reducing viscosity strengthens convection relative to diffusion. Relaxation, smaller physical steps or outer coefficient updates can improve discrete equation convergence. Measure these changes at the same spatial scheme and accuracy target.

C. Physical/model behavior

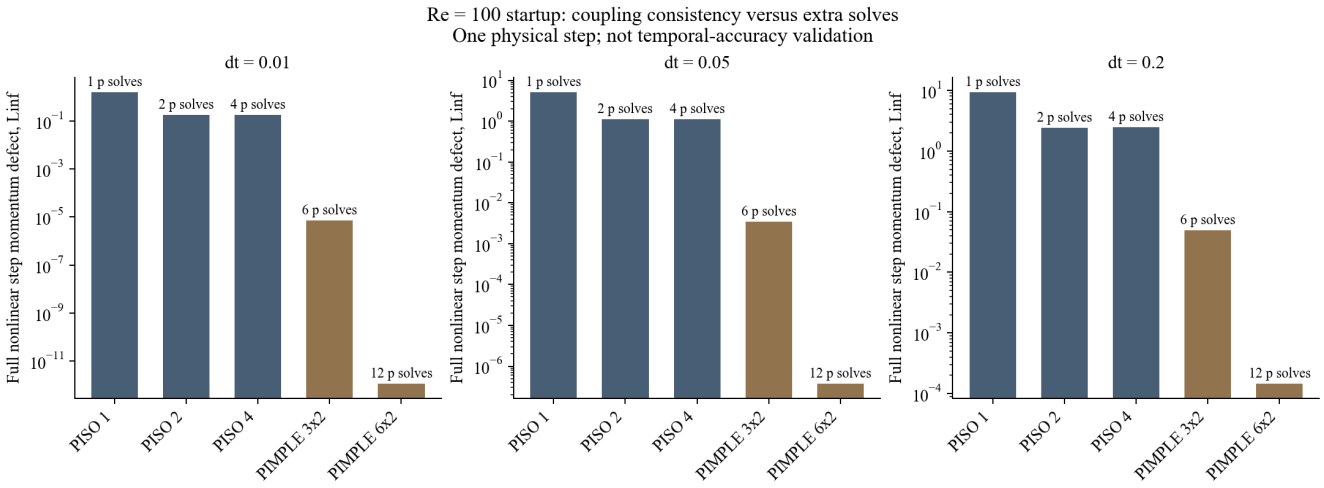
A real higher-Re cavity can develop unsteady or three-dimensional motion. A 2D steady solver does not establish that physical state. Nonconvergence alone does not prove an algorithm bug or the need for a turbulence model.

Albensoeder and Kuhlmann (2005), Accurate three-dimensional lid-driven cavity flow.

Release boundary

The public solver still rejects Re other than 100. Re=400 and 1000 are proposed qualification studies, not newly validated cases. No numerical source or original Week-1 asset was changed for this supplement.

11. A measured within-step PIMPLE advantage



Fifteen configurations start from rest at Re=100 on 32 cells per side. Each takes one backward-Euler step. The full nonlinear step defect rebuilds convection from the final velocity, unlike the frozen PISO predictor defect. Each panel has its own logarithmic y range.

dt=0.05 configuration	Nonlinear Linf	Pressure solves	Momentum solves
PISO 1 inner	5.128e+00	1	2
PISO 2 inner	1.110e+00	2	2
PISO 4 inner	1.123e+00	4	2
PIMPLE 3 x 2	3.434e-03	6	6
PIMPLE 6 x 2	3.750e-07	12	12

All fifteen cell-divergence defects are below 10^{-9}. Extra inner corrections can plateau in the nonlinear defect because the predictor coefficients remain frozen. PIMPLE updates those coefficients, at higher solve cost.

What this establishes

A demonstrated reduction in within-step algebraic error. It is not a physical time-error estimate, proof of an accurate larger dt, or a universal wall-time advantage.

12. Classroom experiment and assessment plan

A. Already executed: common steady state

Re=100, 32/64 cells. Show that all three FV methods reach the same discrete steady field. SIMPLE is cheaper on the coarse grid; PISO is cheaper with the selected fine-grid settings. Keep the nonmonotone FV Ghia errors visible.

B. Already executed: one-step consistency

Re=100, 32 cells; vary inner and outer counts at each fixed dt. Compare the full nonlinear momentum defect, frozen predictor defect, mass balance and solve counts. More pressure corrections and more coefficient updates address different errors.

C. Proposed next: temporal accuracy at Re=100

Compare startup at matched final times; halve dt. Then add a smooth ramped/oscillating lid and a verified temporal reference. Compare field error, energy, amplitude/phase and cost. Use the same time scheme. Transient SIMPLE needs an outer solve inside every physical step; the current steady iteration is not a trajectory.

D. Proposed later: higher-Re qualification

Test Re=400, then 1000 with refined 64/128/256 meshes and independently validated convection. Inspect secondary vortices, centerlines and lid-wall stress. Separate spatial errors from coupling errors and physical unsteadiness.

Assess cost versus error, not a preferred winner. Keep the mesh, linear solver, initial fields and spatial equations fixed for a coupling comparison. Repeat wall-time measurements and retain failed configurations. A dt-refined numerical reference has its own error; document its refinement.

13. Exercises and the next experiment

1. Sum cell divergence and cancel shared interior face terms.
2. Add a pressure constant; verify unchanged velocity.
3. Compare one and two PISO corrections using the one-step momentum defect.
4. Check PIMPLE with one outer loop against PISO.
5. Change SIMPLE relaxation at fixed final residual tolerance.
6. Change only convection interpolation; separate spatial error from coupling effects.
7. Design a Δt-halving startup experiment before claiming a PIMPLE speedup.

Next qualification

Start with a ramped or oscillating lid. Use the same temporal scheme and matched final times. Refine Δt; measure field and phase errors. Any transient streamfunction pressure reconstruction must include acceleration terms.

Only Re=100 steady behavior is qualified in this release. Higher-Re behavior, transient accuracy and stretched grids remain separate tasks.

14. Reading, attribution and reproducibility

PySIMPLE: educational MAC finite-volume SIMPLE

NIST FiPy: pressure-correction derivation (Stokes example)

OpenFOAM: solution and algorithm control

pyOpenFOAM: another inspectable Python loop organization

Ghia, Ghia and Shin (1982): benchmark tables

This independently written module uses original implementation and explanations. External examples inform derivation and organization; their validation claims are not inherited. The audited CFD-Python PISO-titled notebook had a commented-out second correction, illustrating why source inspection matters.

Reproduce and audit

qa/run_week01_2.py executes the steady cases; qa/week01_2_teaching.py executes the one-step sweep and draws flowcharts; qa/build_week01_2.py builds the figures and notebook. tests/test_pressure_velocity.py checks conservation, pressure gauge, PISO behavior, PIMPLE limits and the common steady solution. The expanded teaching supplement provides full source links and study boundaries.

Lecture and code are English. Typography: Times New Roman with vector mathematical notation. Prepared 09 October 2026; equation typography revised 15 October 2026.

Worked example: a conservative two-cell correction

Use two unit-volume cells, impermeable external faces and one shared face velocity q, positive to the right. Unit spacing and d=1 isolate the signs from mesh factors.

$$D_1 = 1, \quad D_2 = -1, \quad G_1 = -1, \quad G_2 = 1$$

For q*=1 the two cell defects are +1 and -1. Their sum vanishes because the shared face enters with opposite signs; neither cell yet satisfies continuity.

$$L = -DdG, \qquad Lp' = -Dq^*$$

Here L has rows (1,-1) and (-1,1). Pin p2'=0. The first row gives p1'=-1; then Gp'=1 and q_new=1-1=0. Adding the same constant to both pressures changes no face gradient. This is the constant pressure nullspace.

Exercise with solution

Change q*=2 and d=0.5. L becomes half the previous matrix. With p2'=0, p1'=-4, Gp'=4 and q_new=2-0.5(4)=0. Applying only alpha_p of the face correction would leave a cell mass defect. In SIMPLE, alpha_p relaxes the pressure update, while the velocity correction remains conservative.

This is a nondimensional sign-and-conservation control, not a resolved cavity. Before using the sparse solver, write the same cancellation for three cells and verify D times a constant internal through-flux at the impermeable ends.